Assignment No. 10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
KD3_rutuja_96226>CREATE PROCEDURE sp_rectangle1(len INT, wid INT)
    -> BEGIN
    ->  DECLARE Area INT;
    ->  DECLARE peri INT;
    ->  SELECT len*wid INTO area;
    ->  SELECT 2*(len + wid) INTO peri;
    ->  SELECT Area, peri AS Perimeter;
    -> END;
    -> $$
Query OK, 0 rows affected (0.052 sec)

KD3_rutuja_96226>DELIMITER ;
KD3_rutuja_96226>
KD3_rutuja_96226>call sp_rectangle1(5,6);
+------+-----------+
| Area | Perimeter |
+------+-----------+
|   30 |        22 |
+------+-----------+
1 row in set (0.019 sec)

Query OK, 0 rows affected (0.051 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD3_rutuja_96226>DELIMITER $$
KD3_rutuja_96226>CREATE PROCEDURE sp_calculate(num INT)
    -> BEGIN
    ->  DECLARE Square INT;
    ->  DECLARE cb INT;
    ->  SELECT num*num INTO Square;
    ->  SELECT num*num*num INTO cb;
    ->  INSERT INTO result VALUES(num, square, cb);
    -> END;
    -> $$
Query OK, 0 rows affected (0.122 sec)

KD3_rutuja_96226>DELIMITER ;
KD3_rutuja_96226>call sp_calculate(8);
Query OK, 1 row affected (0.168 sec)

KD3_rutuja_96226>select * from result;
+------+--------+---------+
| num  | Square | cubeNum |
+------+--------+---------+
|    8 |     64 |     512 |
+------+--------+---------+
1 row in set (0.013 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are

C= (F-32) *5/9 F= 9/5*C + 32

```
KD3_rutuja_96226>DELIMITER $$
KD3_rutuja_96226>CREATE PROCEDURE sp_temperature(type CHAR, value DOUBLE)
    -> BEGIN
    ->  DECLARE cel DOUBLE;
    ->  DECLARE fah DOUBLE;
    ->  IF(type = 'C')
    ->  THEN
    ->  SET fah = 9/5*value + 32;
    ->  END IF;
    ->
    ->  IF(type = 'F')
    ->  THEN
    ->  SET cel = (value - 32)*5/9;
    ->  END IF;
    ->
    ->  SELECT cel AS Celsius, fah AS Fahrenheit;
    -> END;
    -> $$
Query OK, 0 rows affected (0.183 sec)

KD3_rutuja_96226>DELIMITER ;
KD3_rutuja_96226>call sp_temperature('C',15);
+---------+------------+
| Celsius | Fahrenheit |
+---------+------------+
|    NULL |         59 |
+---------+------------+
1 row in set (0.021 sec)

Query OK, 0 rows affected (0.111 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD3_rutuja_96226>DELIMITER $$
KD3_rutuja_96226>CREATE PROCEDURE sp_measure(val INT)
    -> BEGIN
    ->  DECLARE Yard INT;
    ->  DECLARE Feet INT;
    ->  SELECT val/36 INTO Yard;
    ->  SELECT val/12 INTO Feet;
    ->
    ->  SELECT CONCAT(val,' inches equals ',Yard,' yards, ',Feet,' Feet');
    -> END;
    -> $$
Query OK, 0 rows affected (0.154 sec)

KD3_rutuja_96226>DELIMITER ;
KD3_rutuja_96226>call sp_measure(124);
+-------------------------------------------------------------+
| CONCAT(val,' inches equals ',Yard,' yards, ',Feet,' Feet') |
+-------------------------------------------------------------+
| 124 inches equals 3 yards, 10 Feet                          |
+-------------------------------------------------------------+
1 row in set (0.130 sec)

Query OK, 0 rows affected (0.206 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD3_rutuja_96226>CREATE PROCEDURE sp_greaterNumber(num1 INT, num2
 INT)
    -> BEGIN
    ->  DECLARE result INT;
    ->  SET result = num1 * num2;
    ->  IF result > 100
    ->  THEN
    ->  SELECT CONCAT(result, ' is greater than 100');
    ->  ELSE
    ->  SELECT CONCAT(result, ' is less than 100');
    ->  END IF;
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.076 sec)

KD3_rutuja_96226>DELIMITER ;
KD3_rutuja_96226>call sp_greaterNumber(10,11);
+----------------------------------------+
| CONCAT(result, ' is greater than 100') |
+----------------------------------------+
| 110 is greater than 100                |
+----------------------------------------+
1 row in set (0.011 sec)

Query OK, 0 rows affected (0.034 sec)
```